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Unit 1 - Analog vs. Digital

PDF Documents

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UNIT 1 ANALOG VS. DIGITAL



We live in a digital age. As technology has advanced, there has been a shift from analog to digital methods, and now, nearly all information is stored and transmitted digitally. In this digital era, we can process information faster and more accurately, with devices like computers, smartphones, and various digital gadgets becoming central to our daily lives.



What are analog and digital?

Analog and digital are different ways of representing information.

ANALOG

Analog represents information continuously. For example, a thermometer with mercury smoothly rises and falls to show the temperature, illustrating how analog data changes constantly over time. Many natural phenomena, such as sound, light intensity, and temperature changes, are represented analogously.

DIGITAL

Digital represents information by breaking it down into numbers. For example, a digital clock displays time by changing the numbers every second. Digital devices like computers and smartphones convert analog information into numbers, allowing it to be stored accurately and quickly copied.

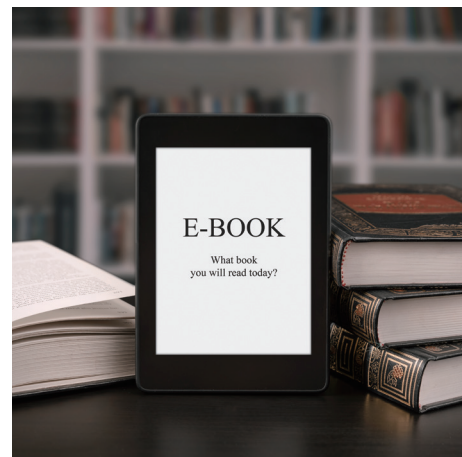


Then, what are some examples of things that have changed from analog to digital?

- 1 CAMERA** Film cameras capture light directly onto film, taking photos in an analog way. In contrast, digital cameras use an image sensor to store photos as digital files. Digital photos are easy to store, share, and edit.



- 2 BOOK** Unlike analog paper books, digital eBooks allow access to many books on a lightweight device. This makes it easy to carry and read numerous books without physical copies.





● Activity: From Analog

ACTIVITY: ANALOG AND DIGITAL

- 1 Think about other examples of things that have changed from analog to digital.

- 2 Write down an analog device that hasn't yet been converted to digital.